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Enhanced speaker-turn aware hierarchical model for automated classroom dialogue act classification

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ABSTRACT

Dialogue act classification (DAC) is a pivotal task in natural language processing, providing deep insights into communication patterns. In classroom settings, analyzing dialogues and delivering timely feedback play a crucial role in improving teaching practices and fostering effective educational interactions. Traditional manual classroom observation is both costly and prone to bias, underscoring the need for automated dialogue analysis. However, recent automated classroom DAC often neglect the importance of speaker turns. To address this gap, we propose the Speaker-Turn Aware Hierarchical Model (SA-DAC), which leverages speaker embeddings to enrich contextual understanding and improve classification accuracy. We further introduce a new benchmark dataset, IRE-9, designed around the initiate-response-evaluate (IRE) encoding framework, to advance research in dialogue analysis. Comprehensive experiments demonstrate that SA-DAC surpasses state-of-the-art methods, delivering superior accuracy and scalability. Moreover, we develop an automated classroom analysis system powered by SA-DAC, which generates detailed, multi-dimensional reports to assist educators in improving their teaching strategies.

1. Introduction

Classroom dialogue is a primary medium of teaching and is closely linked to student academic performance and the productiveness of lessons is influenced by the quality of educational dialogue in whole-class situations (Howe & Abedin, 2013). Traditional classroom dialogue analysis is performed by educational experts and requires extensive training and expertise. As education advances rapidly, manual analysis of classroom conversations can no longer provide timely feedback to teachers. In response, several studies have adopted machine learning methods to automatically classify classroom dialogues, including approaches like Random Forest (Wang et al., 2014), Logistic Regression (Blanchard et al., 2015) and SVM (Xu et al., 2019). However, most research has been limited by small-scale data and lacks a systematic approach. To address these issues, deep learning has been introduced to enhance the capabilities of classroom analysis systems, becoming a focal point of current research. For instance, Suresh et al. (2021, 2019) systematically utilize deep learning models to evaluate K-12 mathematics teacher dialogue and introduce the TalkMoves dataset. Meanwhile, Jensen et al. (2020) presented a method for generating automated feedback on key teacher dialogue variables using self-recorded classroom audio. More recently, Kumaran et al. (2023) proposed SSCon, which employs two independent models leveraging contrastive learning to tackle the few-shot classification in classroom dialogue. Although the aforementioned methods have been successful in classroom dialogue analysis, they overlook the importance of speaker turn dynamics within specific contexts, which has been proved can significant improve DAC performance (He et al., 2021; Shang et al., 2020). As Mercer (2004) argues, a static classification of utterances cannot reflect how meaning is co-constructed through dynamic speaker interactions. Speaker role transitions-such as from teacher to student-often signal shifts in epistemic authority and cognitive scaffolding. This reinforces the need to incorporate speaker dynamics into dialogue act modeling. For example, when a teacher asks a “Question”, the subsequent student response is likely to be categorized as an “Answer”. However, if the same speaker continues, then the following act might be defined as “Self Q&A”. Thus, modeling speaker turns in classroom dialogue is essential for more accurate analysis.

To bridge this gap, we propose a novel classroom dialogue act classification model (SA-DAC). Unlike methods that rely solely on the

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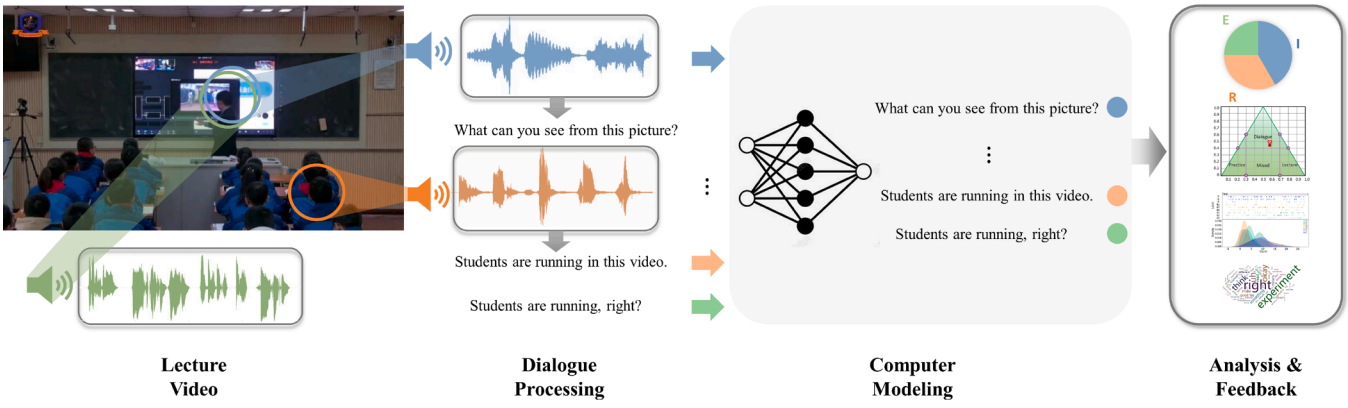


Fig. 1. The overall workflow of our automated classroom analysis system, which starts with classroom recordings and outputs a multidimensional analysis report.

textual content of classroom dialogues, SA-DAC enriches contextual understanding by incorporating speaker embeddings. While previous approaches often use complex models to integrate speaker information, SA-DAC simplifies this process by using global additive embedding vectors based on speaking activity. Additionally, we developed a real-world classroom interaction dataset, *IRE-9*, based on the initiate-response-evaluate (IRE) dialogue encoding framework (Mehan, 1979). Extensive experiments on IRE-9 and other publicly available datasets demonstrate SA-DAC's advantages over state-of-the-art methods and its scalability across diverse datasets. Leveraging SA-DAC, we created an automated classroom analysis system (see Fig. 1) that produces comprehensive, multi-dimensional reports for individual lessons, providing intuitive tools for visualizing and analyzing classroom interactions. Our contributions are summarized as follows:

- We introduce SA-DAC, the first approach to incorporate speaker turns (between teacher and student) in DAC within classroom settings, enhancing classification accuracy and contextual sensitivity.
- We present IRE-9, a novel benchmark dataset based on the initiate-response-evaluate (IRE) framework, advancing research in classroom dialogue analysis with a standardized structure.
- Leveraging SA-DAC, we developed an automated classroom analysis system capable of generating multi-dimensional reports, assisting teachers in refining instructional practices and supporting their professional development.

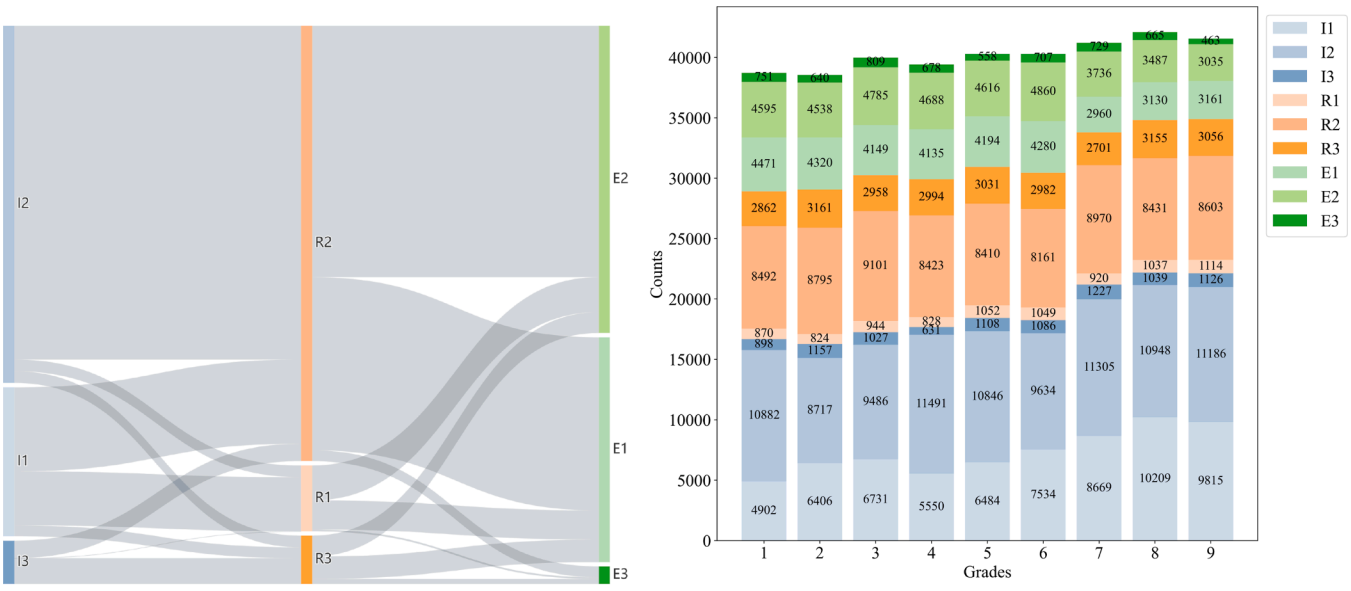
2. Related work

2.1. Classroom dialogue analysis

With the advancement of teaching technology, the academic community's interest in exploring classroom dialogue analysis systems has grown rapidly. Notably, the Flanders (1965) Interaction Analysis System (FIAS) still stands as the most widely adopted foundational theory for dialogue analysis. Sinclair et al. (1975) present a three-step pattern as the paradigm participation structure, characterized as Initiation-Response-Feedback (IRF), with teachers taking primary responsibility for the initiation, students providing responses, and teachers following with feedback. The Academically Productive Talk (APT) framework (Michaels et al., 2008) is designed to enrich classroom discussions, promoting critical thinking and encouraging students to articulate their reasoning effectively. However, existing researches in DAC primarily treat it as a task of classifying individual utterances, often overlook the importance of speaker and contextual information. For instance, Wang et al. (2014) developed a classification model that can distinguish three classroom activities (lecturing, discussion and group work) in mathematics lessons. Xu et al. (2019) employ TF-IDF and Word2vec features, along with a chi-square feature selection method, to classify different teacher dialogues (lecture, instruction, and question). With the powerful representational capabilities of deep learning, Lei et al. (2020) propose a hybrid model, CNN-BiLSTM-Attention, learning local and global features within a seven-class encoding framework. Similarly, Lugini and Litman (2020) incorporate local and speaker information in computational models for argument component classification in classroom discussions. The introduction of the TalkMoves dataset (Suresh et al., 2021), the first standardized compilation of dialogues in K-12 mathematics classes, marked another milestone. More recently, Kumaran et al. (2023) introduce SSCon, a contrastive learning-based self-supervised method for recognizing dialogue acts, demonstrating its effectiveness with limited labeled data. The findings of the aforementioned work highlight the need for DAC systems that can provide timely feedback for teachers. However, the accuracy of classification and the interaction between speakers and dialogue still require further investigation. In parallel with verbal discourse modeling, recent studies have advanced multimodal classroom behavior analysis. For example, the work by Zheng et al. proposes a event segmentation framework using synchronized vision and audio data to detect teaching behaviors and gesture patterns in real time (Zheng et al., 2024). While their approach emphasizes nonverbal cues such as body movement and instructional gestures, SA-DAC exclusively models discourse and speaker patterns, offering a lightweight yet effective alternative for dialogue-centric behavior analysis.

2.2. Dialogue act classification

Early research in DAC primarily approached the task as a text classification problem. Traditional machine learning techniques such as hidden markov models (HMMs) (Stolcke et al., 2000) and conditional random fields (CRFs) (Zimmermann, 2009) were employed to capture sequential dependencies in dialogue, modeling language patterns to identify various dialogue acts. These initial methods often relied on hand-crafted features, which limited their capacity to capture complex linguistic relationships. With the advent of deep learning, a shift occurred, bringing neural network into prominence for DAC, particularly through recurrent neural networks (RNNs) and long short-term memory (LSTM) networks (Lee & Dernoncourt, 2016). For example, Yan et al. (2021) present QP-GAT, which enhances emotion-cause pair extraction and joint dialogue act recognition with sentiment classification, achieving strong performance and providing interpret ability in emotion-cause analysis. The development of attention mechanisms further expanded the capabilities of neural models by allowing them to focus on relevant parts of dialogue context, thus improving DAC accuracy by better capturing contextual and semantic nuances (Ortega & Vu, 2017). More recently, the focus has shifted toward leveraging large-scale pre-trained language models, such as BERT and GPT, which provide rich contextual representations that are particularly beneficial for DAC (Qin et al., 2020, 2021). In parallel, some studies began exploring hierarchical and structural innovations in DAC architectures. A notable study (Mezza et al., 2022) introduced a



(a) Distribution of IRE-9 dialogue triplets. (b) Distribution of IRE-9 dialogues across grades.

Fig. 2. Distribution of IRE-9 dialogue patterns in the dataset.

neural architecture that models hierarchical label dependencies, incorporating both contextual and semantic features. Additionally, Xu et al. (2022) propose an end-to-end graph attention reasoning network for conversation understanding that captures long-term contextual dependencies at both word and utterance levels. To enhance sequence consistency in DAC predictions, some researchers have also integrated CRFs as a post-processing layer in neural architectures, effectively enforcing coherence across dialogue label sequences (Kumar et al., 2018; Li et al., 2019; Shang et al., 2023). Beyond these improvements in modeling techniques, recent research highlights the potential of incorporating speaker information to enhance classification accuracy by leveraging turn-taking patterns and conversational roles (He et al., 2021; Zhao et al., 2022). Despite these advancements, there remains a gap in the application of these techniques specifically to classroom dialogue classification. To our knowledge, no existing work has applied advanced modules, including hierarchical structures and speaker information, within the educational context. Thus, our study aims to bridge this gap by investigating how these elements can be optimally utilized for DAC in classroom environments, adapting and refining these methods to support effective classroom dialogue analysis.

3. IRE-9 classroom dialogue dataset

3.1. Data collection

The *IRE-9* was obtained from daily classroom recordings, secured with prior informed consent for the use of this de-identified dataset for research purposes. Our dataset covers a diverse range of subjects and grade levels, spanning from primary to middle school. The distribution of dialogues across these grades are illustrated in Fig. 2. The dataset comprises approximately 1008 classroom recordings, with each grade from 1 to 9 contributing 112 lessons. On average, each lesson contains around 400 valid utterances after ASR transcription. An interesting trend observed in Fig. 2(b), shows that as grade levels increase, there is a stronger emphasis on teacher-led instruction (I). Additionally, there is a notable imbalance in the number of labels within the classroom data, which warrants attention. Our dataset mainly comprises Chinese language dialogues, with some English lesson conversations included. For the ASR task, we use WhisperX[1] (Bain et al., 2023). The speaker diarization module in WhisperX is powered by the pre-trained `pyannote/speaker-diarization-3.1` model[2], which performs speaker turn detection based on audio segmentation and clustering. However, the classroom transcriptions still contained some noise, including filler sounds like "uh" and recognition errors.

To address these issues and clean the data, we leverage the ChatGPT API[3] to rewrite sentences, misspellings and missing words. This approach helped standardize the data, making the sequence classification task more manageable. We adopt the rewriting strategy introduced in our prior work (Jia et al., 2025), which applies prompts only to localized ASR errors while preserving utterance intent. Full details, including prompt design and example outputs, are available in that study.

3.2. Dialogue annotation

Building on the theoretical foundations and automated dialogue analysis proposed by Yang et al. (2023), we expanded the IRE, allowing us to thoroughly analyze verbal interactions, which are sufficient for determining the overall teaching type in most classroom settings. To be more specific, we invited experts specializing in the field of the *"Theory of Curriculum and Instruction"* to create a novel coding framework – *IRE-9*, which comprises distinct hierarchical structures. The first category is divided into three categories “I, R, E”, each of which further includes three sub-levels, as illustrated in Table 1. To ensure the quality of the labeling process, we conducted a Kappa (McHugh, 2012) analysis to assess the inter-rater reliability among 20 annotators involved in labeling our dataset. The Kappa coefficients were calculated at two different points: initially and after multiple sessions of annotation. The initial Kappa coefficients ranged from 0.82 to 0.90, and the midpoint Kappa coefficients ranged from 0.87 to 0.96, indicating that the annotators achieved a substantial to almost perfect level of agreement as the labeling process progressed (see Table 2). This improvement suggests that repeated sessions allowed annotators to better align on coding decisions, resulting in more consistent annotations over time.

[1] https://github.com/m-bain/whisperX
[2] https://huggingface.co/pyannote/speaker-diarization-3.1
[3] https://openai.com/chatgpt/

Table 1
Definitions and example utterances for *IRE-9* coding framework.

Category	Definition	Example
I_1	Closed questions with standard answers.	"Is it right?"
I_2	Explanatory questions with standard answers.	"What can you see from this picture?"
I_3	Open-ended questions without standard answers.	"What have you learned from this lesson?"
R_1	Simple response.	Yes/ No
R_2	Factual knowledge answer	"Students are running in this video."
R_3	Open or explanatory answer.	"I have learned how to solve algebraic equations."
E_1	Simple reply or direct evaluation	"Students are running, right?"
E_2	Retell the respondent's point of view, and invite the respondent to further explain.	"Have you made any other discoveries in this picture?"
E_3	Invite others to evaluate and answer, promote reflection and group discussion.	"Is there anyone else who can help him answer this question?"

Table 2
Cohen's kappa scores between annotators.

Coding decision	Inter-rater agreement	Initial kappa	Midpoint kappa
I_1	95 %	0.88	0.93
I_2	94 %	0.90	0.94
I_3	85 %	0.84	0.88
R_1	89 %	0.82	0.87
R_2	90 %	0.88	0.96
R_3	88 %	0.87	0.89
E_1	92 %	0.90	0.93
E_2	90 %	0.87	0.91
E_3	86 %	0.85	0.87

3.3. Annotation explanation

In classroom dialogue analysis, IRE stands for Initiation-Response-Evaluation. This pattern involves a teacher initiating a discussion, a student responding, and the teacher evaluating the student's response. As shown in Table 1, the first level has three classes of labels, namely I, R, and E. The second level has nine classes of labels, with each of the first level categories being further divided into three subcategories (i.e. I_1, I_2, I_3). The second level represents a progression from basic to advanced dialogue within the current category. For instance, I_1 involves simple teacher questioning, while I_3 encourages students to use creativity, critical thinking to participate in answering.

It's important to note that a single dialogue can convey different meanings within different contexts. For instance, when a teacher poses a question and immediately provides the answer, this dialogue lacks interaction between the teacher and student. This utterance would be labeled as "I". Conversely, the same dialogue may be categorized as "E" if it serves as follow-up questions in response to students' answers. These subtle distinctions emphasize the significance of examining speaker turns within the context of the dialogue.

4. Method

In this section, we present an overview of the design and structure of our SA-DAC model, detailing its core components and underlying mechanisms as illustrated in Fig. 3.

4.1. Task definition

In dialogue analysis, **speaker turn** refers to the period when a single speaker actively speaks in a conversation, beginning when the speaker starts talking and ending upon pausing, concluding a thought, or being interrupted by another speaker. Analyzing speaker turns, particularly in educational contexts, is crucial for understanding the flow of interaction and uncovering patterns in exchanges, such as those between teachers and students.

The task of DAC is defined as follows: Given a conversation sequence $D = (U_n, Y_n, S_n)_{n=1}^{N}$, each D represents a dialogue with sequences of varying lengths, where $U_n = \{u_1, u_2, \ldots, u_n\}$ stands for a dialogue instance. Additionally, $Y_n = \{y_1, y_2, \ldots, y_n\}$ represents the ground truth labels corresponding to each utterance, and $S_n = \{s_1, s_2, \ldots, s_n\}$ comprises speaker labels which composed of *Teacher*, *Student* and *Group activity*. For instance, consider the utterance, *"What can you see from this picture?"* has the input format $\{u, I_2, Teacher\}$. The goal is to learn a classification

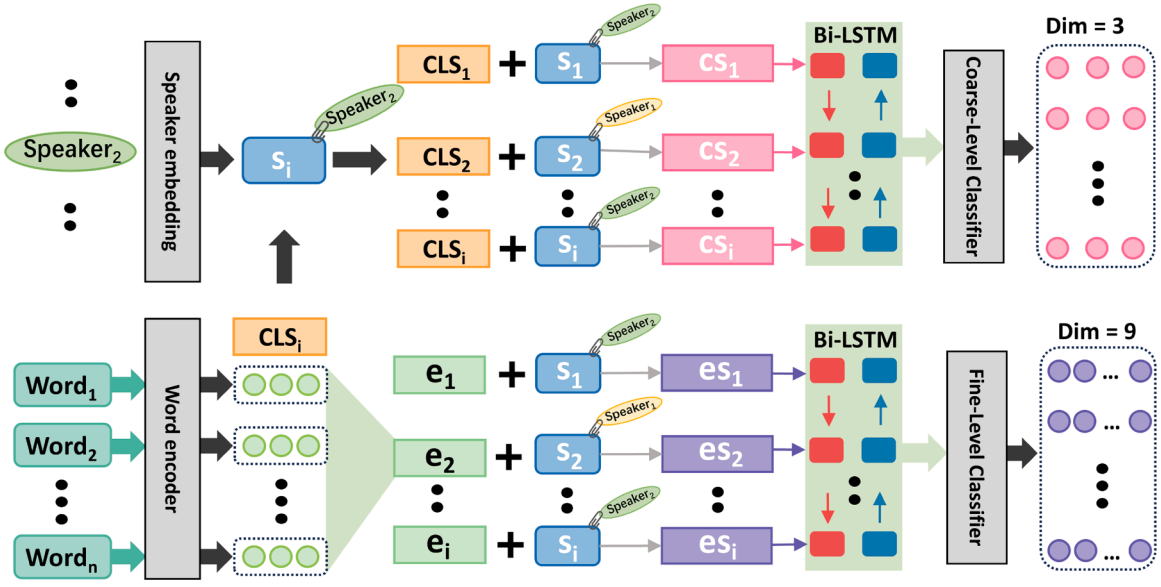


Fig. 3. Architecture of the hierarchical classroom dialogue classification model.

model that can accurately extract speaker turns from the dataset and precisely categorize each utterance.

4.2. Utterance modeling

As shown in Fig. 3, the input to the model consists of a series of utterances, each accompanied by its corresponding speaker ID. To effectively capture both the linguistic content and the speaker-specific characteristics of these utterances, our approach employs a two-layer embedding strategy: a speaker embedding and a word encoder. Each utterance, along with its speaker information, undergoes an embedding lookup process in the speaker embedding layer to obtain the corresponding speaker ID embedding, which is then combined with the word embedding. By incorporating speaker information, we aim to model the influence of the speaker's identity on the dialogue, which is particularly important for capturing the dynamics of classroom conversations. Following the speaker embedding, each utterance undergoes a word encoding process using a pre-trained language model – RoBERTa (Liu et al., 2019). The output from the word encoder is a sequence of context-aware word embeddings. The representation of each utterance U_i is given by:

$$U_i = [\mathrm{CLS}_i], w_1, w_2, \ldots, w_n, [\mathrm{SEP}] \tag{1}$$

where U_i denotes the ith utterance, and $w_1, w_2, \ldots, w_n$ are the words in the utterance. The [CLS] token is a special token used for aggregating the information of the entire utterance, while the [SEP] token signifies the end of the utterance, aiding in the delineation of different segments within the dialogue.

4.3. Speaker turn modeling

The encoded word representations are then processed to generate contextual embeddings. As depicted in the Fig. 3, the output from the word encoder is a series of embeddings for each word, with the [CLS] token playing a crucial role in summarizing the utterance-level information. This approach ensures that our model can effectively capture both the individual meaning of each word and the overall semantics of the utterance within a speaker-aware context, contributing to a robust and contextually informed representation of classroom dialogues.

Unlike monologic texts authored in non-interactive settings, classroom dialogues involve multiple participants whose turns signal dynamic shifts in communicative roles. It is therefore crucial to model speaker transitions explicitly. This design is informed by Mercer's Sociocultural Discourse Analysis (Mercer, 2004), which emphasizes that speaker transitions in classroom discourse often correspond to shifts in epistemic authority and instructional function. By incorporating role-sensitive speaker identification, our model captures this underlying pedagogical structure, allowing it to differentiate between utterances not only by content but also by their position within the instructional dialogue. To address this, we introduce three speaker embeddings, Teacher, Student, and Group activities, which are independent of any specific conversation. Similar to the positional encoding in Transformers (Vaswani et al., 2023), these speaker embeddings are learnable parameters optimized during training and share the same dimensions as the utterance embeddings. The output from WhisperX provides timestamps, speaker labels, and utterances. To initially identify the teacher's speech, we performed the following operations:

$$(T, S, U) = \text{WhisperX (Input Audio)} \tag{2}$$

where T represents the set of timestamps, S the set of speaker labels, and U the set of utterances. For each speaker label $s_i \in S$, we calculated the total duration d_i and frequency F_i of each speaker label. Then, teacher's voice was identified by selecting the label with the longest duration and highest frequency:

$$s_{\text{teacher}} = \arg\max_{s_i}(d_i + F_i) \tag{3}$$

Subsequently, we identified group activities (noisy) and student contributions (diverse and shorter in duration). For example, if the initial sequence is [0, 0, 1, 0, 0, 2], it can be relabeled as [T, T, S, T, T, S], where 'T' represents the teacher and 'S' represents the student. When the label changes, it indicates that a speaker turn has occurred. We summarize the speaker label identification process, including the teacher label assignment based on duration and frequency, in Algorithm 1.

Algorithm 1 Speaker label identification from WhisperX.

Input: Audio segment
Output: Teacher label s_{teacher}

(T, S, U) = WhisperX(audio)
Initialize empty dictionary: score = {}
for each speaker label s_i in S **do**
    duration $d_i \leftarrow$ total speaking time of s_i
    frequency $F_i \leftarrow$ number of utterances of s_i
    score$[s_i] = d_i + F_i$
end for
$s_{\text{teacher}} \leftarrow$ speaker label with highest score
Relabel S: assign 'T' to s_{teacher}, 'S' to others

To assess the reliability of our speaker role assignment, we manually annotated a subset of 200 dialogue segments and compared them with the output of our rule-based method. The precision for identifying teacher utterances reached 94.2 %, while individual student utterances achieved 90.5 %. Group discussion segments, due to their overlapping and fragmented speech patterns, were more challenging to classify accurately, yielding a precision of 84.3 %. Most of the errors in this category resulted from mislabeling group interactions as student turns. Despite these limitations, the overall accuracy of the speaker role assignment strategy is sufficient to support downstream dialogue act classification.

4.4. Hierarchical dialogue classification

Instead of relying solely on the *[CLS]* token, we further process the word embeddings generated by the encoder. After encoding the utterances, we obtain the outputs e_i and $[CLS_i]$. To capture more detailed information, we then concatenate these outputs with the speaker embeddings.

$$es_i = [e_i, s_i] \tag{4}$$

$$cs_i = [CLS, s_i] \tag{5}$$

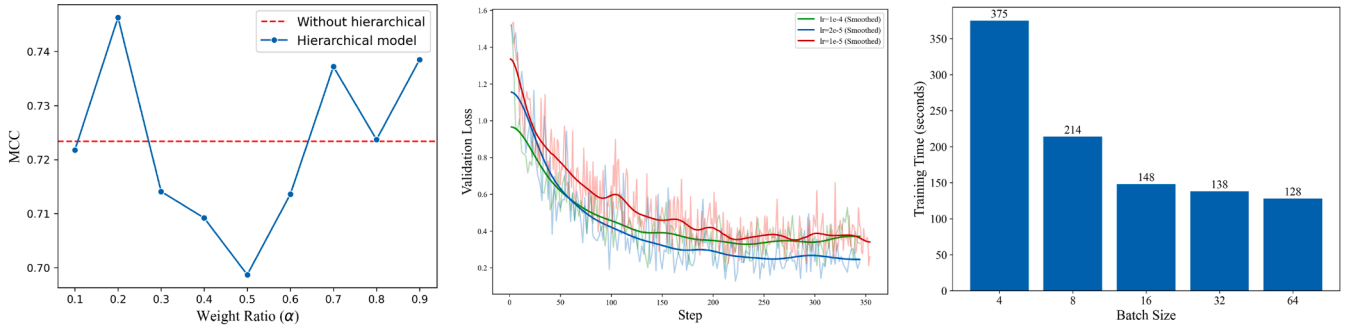
We further apply a Bidirectional Long Short Term Memory (Bi-LSTM) to process these embeddings.

$$\overrightarrow{h_{it}} = \overrightarrow{\mathrm{LSTM}}(\boldsymbol{es}_i), \overleftarrow{h_{it}} = \overleftarrow{\mathrm{LSTM}}(\boldsymbol{es}_i) \tag{6}$$

where $w_{it} \in H_i$, represents the t-th word embedding of utterance U_i. We concat the bi-direction embedding together to get $\boldsymbol{h}_{it} = [\overrightarrow{h_{it}}, \overleftarrow{h_{it}}]$, which summarizes the context information of w_{it}.

Our model incorporates a hierarchical structure that considers classification at two levels. The first level performs a three-class classification (Dim = 3), while the second level refines these predictions into nine finer-grained categories (Dim = 9). This hierarchical setup allows for a more detailed understanding of dialogue, as each layer captures context-specific information at different levels of granularity. To optimize our hierarchical model, we apply loss functions at each level. Specifically, we combine cross-entropy loss and a CRF layer, with the losses weighted to form the final objective. When training the CRF model, our objective is to find the optimal parameters Θ (comprising both transition and emission parameters), that minimize the loss function:

$$\mathcal{L} = -\alpha\left(\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}\log\hat{y}_{i,k}\right) - \beta\left(\sum_{i=1}^{N}\log P(Y_i \mid C_i, \Theta)\right) \tag{7}$$



(a) Impact of different weight ratio on final MCC. (b) Impact of learning rate on validation loss. (c) Impact of different batch size on training time.

Fig. 4. Impact of different hyper-parameters setting.

Table 3
The statistics of the three datasets. |C| denotes the number of dialogue act (DA) classes; |P| denotes the number of parties; Train/Val/Test denotes the number of utterances in the corresponding split.

Dataset	|C|	|P|	Train	Val	Test
TalkMoves	6	2	10K/143K	2K/42K	2K/42K
DailyDialog	5	2	11K/87.1K	1K/8K	1K/7.7K
IRE-9	9	2	30K/271K	4K/45K	4K/45K

where N is the number of samples, K is the number of classes, $y_{i,k}$ represents the true label of sample i for class k, and $\hat{y}_{i,k}$ is the predicted probability of sample i for class k. $P(Y_i \mid C_i, \Theta)$ is the probability of the label sequence Y_i given the input sequence C_i. In this formulation, α, β are the weights assigned to the cross-entropy and CRF loss components, respectively. The first term represents the average cross-entropy loss across all samples and classes, while the second term captures the log-likelihood of the CRF model. The impact of different α and β ratios on the experimental results is shown in Fig. 4(a).

5. Experiment

In this section, we present experiments conducted on the specified dataset and baselines to evaluate the effectiveness of our approach. These experiments are designed to address the following research questions (RQs):

RQ1: *How effectively does our proposed method classify interactions in the IRE-9 dataset, and can it be practically applied in real-world settings?*
RQ2: *How does our method compare with other state-of-the-art DAC methods in classroom settings, and how scalable is it across multiple datasets?*
RQ3: *How can each component contribute to improving the classification of classroom dialogues?*

5.1. Experiment settings

Datasets: We present an overview of each dataset used in our study, along with key statistics (see Table 3).

We evaluate our model using three datasets: TalkMoves, DailyDialog, and our collected dataset–IRE-9.

- The **TalkMoves** (Suresh et al., 2019) consists of K-12 mathematics lesson transcripts annotated with teacher and student talk moves, containing 143,829 teacher utterances and 42,857 student utterances.
- The **DailyDialog** (Li et al., 2017) is a multi-turn dialogue dataset featuring everyday conversations, categorized into four dialogue acts: "inform" (45 %), "questions" (29 %), "directives" (17 %), and "commissive" (9 %).
- The **IRE-9**, collected and manually annotated from real classroom scenarios, encompasses teacher-student conversations across multiple subjects from grades one to nine. In this study, we treat it as a multi-turn dataset with nine distinct labels. Further details on the dataset can be found in Section 3.

Experimental Setup: Our model was trained using an NVIDIA RTX 4090 GPU with PyTorch, setting the tokenizer length to 128 for input sequences. We employed a pre-trained RoBERTa model with a feature extraction dimension of 1024, followed by a two-layer Bi-LSTM encoder with a hidden state size of 512 to generate contextualized representations of text sequences. For loss function fusion, a predefined hierarchical weighting was applied with $\lambda = 0.2$ and $\gamma = 0.8$, ensuring a total sum of 1. The model was optimized using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 32. As illustrated in Fig. 4, different parameter settings significantly influenced model performance. Specifically, Fig. 4(b) demonstrates the impact of varying learning rates on model training, while Fig. 4(c) shows how different batch sizes affected the training time per epoch.

Evaluation Metrics: To evaluate our model's performance, we employed multi-class classification metrics, focusing on precision, recall, and F1 score. Precision measures the proportion of correctly predicted positives among all positive predictions, while recall reflects the model's ability to capture actual positives. The F1 score, the harmonic mean of precision and recall, balances both false positives and false negatives. These metrics offer a comprehensive view of model performance across all classes. To account for class imbalance, we used macro averages, ensuring a fair evaluation regardless of class distribution. Additionally, we included the Matthews Correlation Coefficient (MCC) (Matthews, 1975), which is robust to imbalanced datasets. MCC ranges from -1 to 1, with higher values indicating better performance, and provides a stable measure of model performance, particularly in imbalanced scenarios (Chicco & Jurman, 2020).

Baselines: We evaluated several deep learning-based DAC models as baselines, encompassing both traditional methods and state-of-the-art approaches. These models include BERT-based architectures, hybrid frameworks, and large language models (LLMs). Before detailing the specific baselines, we briefly describe how multilingual input from IRE-9 is handled across different models. The IRE-9 dataset contains both English and Chinese utterances. While the dataset is multilingual, each individual session is monolingual. To ensure consistent preprocessing across models, we adopted a unified tokenization strategy. All Transformer-based models use their default subword tokenizers. For multilingual-compatible architectures such as BERT and RoBERTa, we employed pretrained models like BERT-base-Chinese[4] and RoBERTa-large[5], whose vocabularies were constructed or fine-tuned on Chinese corpora and are capable of handling both Chinese and English. For GPT-based models (GPT-3.5-turbo and GPT-4o), tokenization relies on the

[4] https://huggingface.co/google-bert/bert-base-chinese
[5] https://huggingface.co/FacebookAI/xlm-roberta-large

Table 4
Normalized confusion matrix between coarse-level I/R/E categories (per row).

Ground Truth	Predicted I	Predicted R	Predicted E
Initiation (I)	**0.92**	0.01	0.07
Response (R)	0.01	**0.95**	0.02
Evaluation (E)	0.09	0.02	**0.89**

multilingual tokenizer embedded in each model architecture. Below, we provide an overview of the baselines used in our experiments:

- **RoBERTa-large** (Suresh et al., 2021): A robust BERT-based model with enhanced pre-training through dynamic masking. It has demonstrated strong performance on benchmark datasets such as Talk-Moves and IRE-9.
- **Distill-BERT** (Sanh et al., 2020): A distilled version of BERT that strikes a balance between computational efficiency and accuracy, making it suitable for large-scale dialogue tasks.
- **RoBERTa-LSTM** (Lei et al., 2020): A hybrid model combining the contextual representation power of RoBERTa with the sequential modeling capabilities of LSTM, effectively capturing conversational flow.
- **DAH-CRF** (Shang et al., 2023): Incorporates a Conditional Random Field (CRF) layer as a post-processing step to enforce consistency in sequence labeling, which is particularly useful for maintaining coherence in dialogue act predictions.
- **GPT-3.5-turbo** (Brown et al., 2020): A widely adopted version of the GPT-3.5 series, known for its high efficiency and effectiveness in zero-shot applications.
- **GPT-4o** (Achiam et al., 2023): The latest release of GPT-4 used in our study, configured consistently across tasks to ensure reliable comparisons.

5.2. Results

5.2.1. Performance on *IRE-9* (RQ1)

To analyze error cascades in our hierarchical classification, we provide a coarse-level confusion matrix (Table 4). Interestingly, due to the integration of role-sensitive speaker embeddings, misclassifications between teacher-led (I/E) and student-led (R) utterances are rare. This indicates that the model has learned to distinguish speaker roles, and coarse-grained misclassification rarely crosses epistemic boundaries (i.e., teacher vs. student).

Table 5
Model performance for each category of *IRE-9*.

Categories	Precision (↑)	Recall (↑)	F1-score (↑)
I_1	0.81 ± 0.01	0.90 ± 0.01	0.86 ± 0.01
I_2	0.87 ± 0.01	0.91 ± 0.01	0.89 ± 0.01
I_3	0.72 ± 0.02	0.61 ± 0.03	0.68 ± 0.02
R_1	0.84 ± 0.01	0.64 ± 0.03	0.65 ± 0.03
R_2	0.83 ± 0.01	0.84 ± 0.01	0.86 ± 0.01
R_3	0.70 ± 0.02	0.61 ± 0.03	0.66 ± 0.02
E_1	0.94 ± 0.01	0.84 ± 0.01	0.89 ± 0.01
E_2	0.70 ± 0.02	0.69 ± 0.02	0.70 ± 0.02
E_3	0.66 ± 0.02	0.59 ± 0.02	0.63 ± 0.02
Macro avg	**0.79 ± 0.01**	**0.73 ± 0.02**	**0.76 ± 0.02**
Weighted avg	**0.82 ± 0.01**	**0.81 ± 0.01**	**0.81 ± 0.01**

To address RQ1, we evaluated the performance of our DAC experiments over multiple runs, with the mean and standard deviation reported in Table 5. The model achieved varying levels of precision, recall, and F1-scores across different classes. Notably, Classes I_1, I_2, R_2, and E_1 demonstrated high precision and recall, leading to strong F1-scores. Fig. 5 show that the model excelled in correctly identifying samples from Class 1, 2, 5, and 7, as indicated by the high recall values in these classes. However, the model encountered difficulties with classes I_2 and I_3, particularly in correctly identifying open questions and more complex interactions. This indicates that the model still requires improvement in its semantic understanding of "open questions" and "closed questions." We observed frequent confusion among the categories R_2, R_3, E_2, and E_3, largely due to overlapping pragmatic functions and the lack of sufficient contextual cues. These categories share similar surface forms or communicative intents, making them challenging to distinguish-especially in short or context-independent utterances. For example, the utterance "I think the water boils at 100 degrees." was labeled as R_2, but was misclassified as R_3. Conversely, "I have learned that different shapes affect how things fall." was labeled as R_3, yet misclassified as R_2. In both cases, the presence of reflective phrases such as "I think" or "I have learned" blurs the line between stating a fact and offering an explanation. These examples illustrate that the semantic boundaries between certain categories can be subtle and context-dependent. Incorporating broader discourse context may help disambiguate such cases and improve classification performance in future work.

5.2.2. Comparison with baselines (RQ2)

To evaluate our proposed method for the DAC task, we performed 5 runs with different random seeds. After running each experiment, we computed the mean and standard deviation for each metric. For

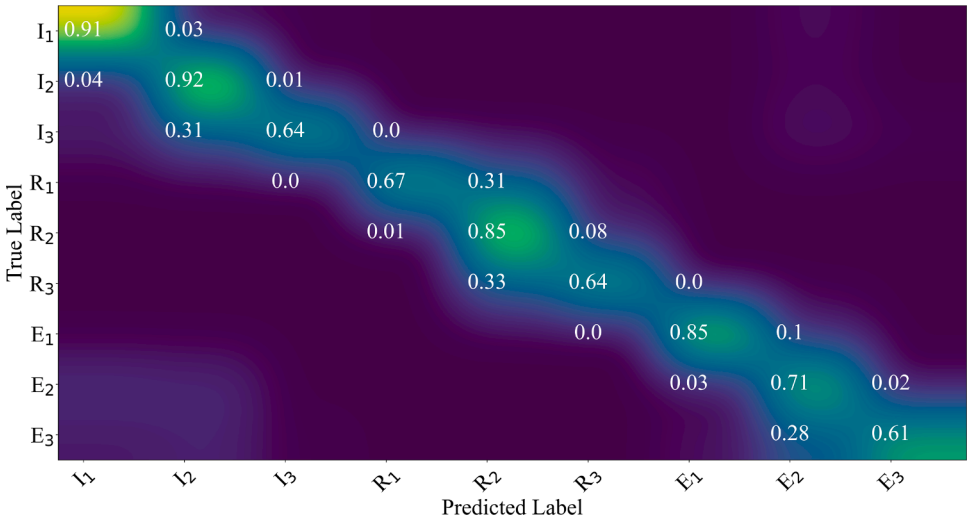


Fig. 5. Confusion matrix of SA-DAC for the IRE-9 dataset (9 DA class labels).

Table 6
Comparison results for the TalkMoves, DailyDialog and IRE-9 dataset with other baselines.

Dataset	Language	Model	Precision (↑) (macro)	Recall (↑) (macro)	F1-score (↑) (macro)	MCC (↑)
TalkMoves	English	BASE-MODEL (Suresh et al., 2019)	–	–	0.65	–
		RoBERTa-large (Suresh et al., 2021)	–	–	0.7933	**0.7779**
		Distill-BERT (Sanh et al., 2020)	–	–	0.7802	0.7616
		Ours	**0.7842 ± 0.0032**	**0.7823 ± 0.0040**	**0.7976 ± 0.0021**	0.7712 ± 0.0045
DailyDialog	English	BERT-base (Devlin et al., 2019)	0.7415	0.7552	0.7420	0.7276
		Co-GAT (Qin et al., 2021)	0.81	0.781	0.794	-
		Ours	**0.8212 ± 0.0047**	**0.8082 ± 0.0051**	**0.8135 ± 0.0036**	**0.7615 ± 0.0035**
IRE-9	English or Chinese (monolingual per session)	BERT-base (Devlin et al., 2019)	0.7018 ± 0.0056	0.6345 ± 0.0065	0.6551 ± 0.0052	0.6429 ± 0.0048
		RoBERTa-LSTM (Lei et al., 2020)	0.7215 ± 0.0049	0.6793 ± 0.0052	0.6985 ± 0.0046	0.6810 ± 0.0037
		RoBERTa-large (Suresh et al., 2021)	0.7432 ± 0.0031	0.7239 ± 0.0036	0.7301 ± 0.0032	0.7249 ± 0.0030
		DAH-CRF (Li et al., 2019)	0.7524 ± 0.0040	0.7289 ± 0.0044	0.7419 ± 0.0038	0.7337 ± 0.0040
		GPT-3.5-turbo (Brown et al., 2020)	0.6068 ± 0.0065	0.5737 ± 0.0070	0.6023 ± 0.0062	0.5975 ± 0.0056
		GPT-4o (Achiam et al., 2023)	0.6502 ± 0.0051	0.6202 ± 0.0058	0.6283 ± 0.0053	0.6222 ± 0.0050
		Ours	**0.7928 ± 0.0037**	**0.7421 ± 0.0041**	**0.7629 ± 0.0035**	**0.7425 ± 0.0038**

baseline methods reproduced from other papers, we have kept the reported results consistent with those originally published. The results are presented in Table 6. For the **TalkMoves** dataset, we focused specifically on the teacher utterances. Since the original dataset lacks frequent speaker role transitions between teacher and student, as well as a hierarchical labeling structure, we primarily focused on assessing the performance of other components of our model. Even without leveraging speaker turn information, our model outperformed existing baselines, demonstrating the effectiveness of our method. On the **DailyDialog** dataset, we observed a slight performance gain after applying speaker turn modeling. However, due to the predictable speaker pattern in this dataset, where conversations typically follow a [0, 1, 0, 1, 0, 1] sequence, the addition of speaker turn information provided limited auxiliary benefits. This highlights the more significant impact of our approach in scenarios where speaker roles are less predictable. On the **IRE-9** dataset, our proposed method demonstrated the best performance across most metrics, highlighting the advantages of incorporating speaker embeddings to enrich contextual information and the effectiveness of our hierarchical classification solution. Unlike other pre-trained language models (PLMs) that ignore role information and treat each sentence as an isolated unit, our approach captures the dynamic interplay among dialogue participants. Additionally, we compared our method against the DAH-CRF (Li et al., 2019), which utilizes CRF and hierarchical structures but does not incorporate speaker information. Our method outperformed this baseline, further validating the importance of modeling speaker roles. To further evaluate our model, we tested it against large language models (LLMs) using the ChatGPT, employing zero-shot learning with GPT models, where only the IRE-9 classification scheme was provided to the model. Our method outperformed the GPT models, particularly in contexts closely aligned with the objectives of our study, where general-purpose GPT models tend to be less effective.

5.2.3. Ablation study (RQ3)

To assess the contribution of various components in our model, we conducted ablation studies. The results in Table 7 show the performance impact when specific components are removed.

Table 7
Impact of key components on model performance.

Model	F1-score (↑)	MCC(↑)
Without Bi-LSTM	0.7558	0.7389
Without CRF	0.7624	0.7426
Without Hierarchical	0.7404	0.7277
Without Speaker-embedding	0.7474	0.7340
Without Speaker Turns	0.7326	0.7210
Full model	**0.7664**	**0.7463**

Without Bi-LSTM: In this variant, the Bi-LSTM is replaced with a unidirectional LSTM to test the significance of bidirectional modeling. While it is generally assumed that in dialogue classification, the subsequent labels are primarily determined by preceding context, our experiments emphasize the importance of considering both preceding and succeeding context for a more accurate classification.

Without CRF: In this setting, the CRF layer is replaced with a multiclass cross-entropy loss function, which we consistently applied across all our experiments. Although prior work (Shang et al., 2020) suggests that the effectiveness of the CRF layer may vary depending on the specific dataset and task, our experiments confirmed this observation. In our case, the inclusion of the CRF layer resulted in only a slight improvement, indicating that its positive impact was limited. Therefore, the decision to use CRF should be carefully considered based on the particular requirements of the task and dataset.

Without Hierarchical: In this configuration, the task is treated as a straightforward 9-class classification, without utilizing the [CLS]. This simplification negatively impacts the model's performance, highlighting the crucial role of the hierarchical structure in accurately understanding and classifying the complex dynamics of classroom interactions.

Without Speaker-embedding: In this variant, the speaker-embedding is omitted, enabling us to evaluate the model's capability to consider contextual relationships without explicitly incorporating speaker information. The results clearly highlight the crucial role of integrating both utterances and speaker details.

Without Speaker Turns: In this ablation variant, we randomly shuffle all utterances within each session, thereby removing speaker turn continuity and conversational flow. This disruption eliminates both speaker adjacency patterns and contextual dependencies, forcing the model to operate on isolated utterances as a sentence-level classifier.

As shown in Table 6, our model built on RoBERTa consistently outperforms the BERT-base and other baselines. This demonstrates the effectiveness of using a strong transformer-based language model. Given that transformer architectures have been widely validated in previous NLP research for their ability to capture long-range dependencies and contextual semantics, we did not conduct an additional ablation study replacing the transformer with simple word embeddings. However, the significant performance differences observed between BERT-base and RoBERTa-based models in our results provide indirect evidence of the transformer's contribution. We explicitly highlight this to emphasize that the improved performance of our model is built upon the strength of transformer representations, further enhanced by our speaker-aware hierarchical design.

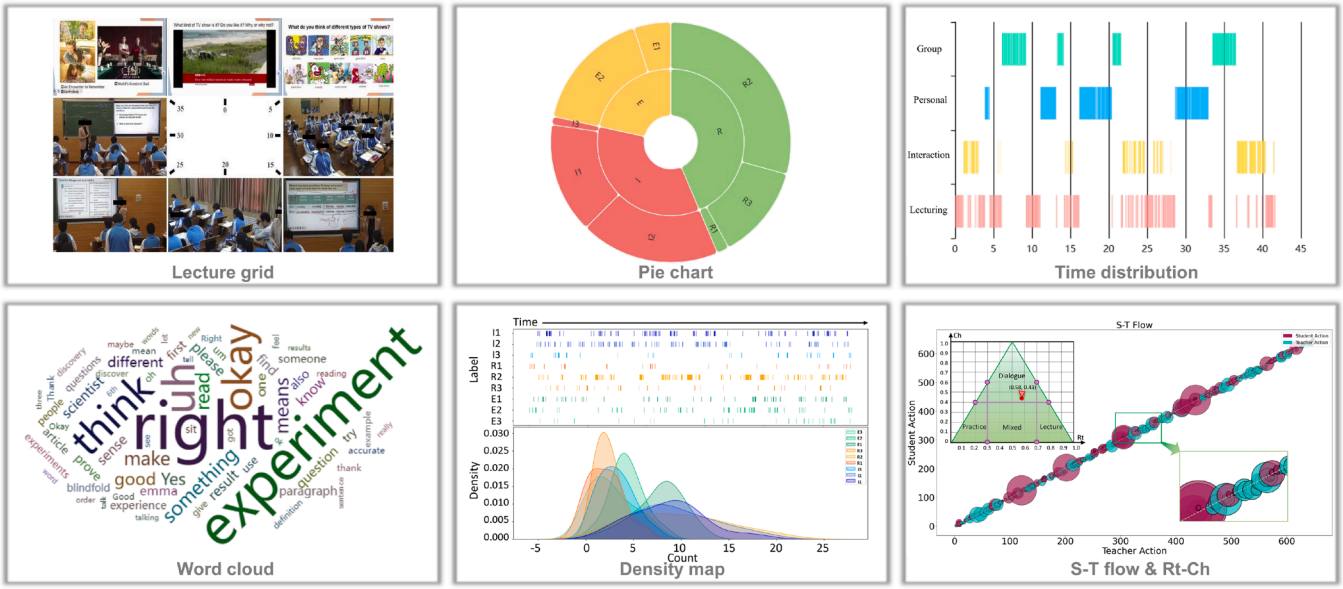


Fig. 6. Screenshot of the analysis system interface for a single classroom recording.

6. System

6.1. System design

The final step in developing the classroom DAC application was to define the feedback that teachers would receive. Using a specific English literature lesson as an example, Fig. 6 illustrates the dashboard generated by the system. The dashboard includes: (1) a classroom temporal grid, (2) the distribution of classroom dialogues, (3) the time distribution of classroom activities, (4) a word cloud, (5) a density map of IRE-9 dialogue types along with kernel density estimation (KDE) curves for each type, and (6) the dialogue flow between students and the teacher, visualized using the Rt-Ch diagram based on the S-T analysis framework (Jun & Huanghai, 2011) for modeling classroom instructional dynamics. To facilitate the interpretation of the S-T & Rt-Ch plot, we clarify that this figure demonstrates the process of dialogue switching between a teacher and students during a class, the green circle represents teacher action, and its size is proportional to the duration of the utterance, while the red represents student action. The upper left is the Rt-Ch figure in which the x-axis corresponds to the Rt value, reflecting the fraction of teacher involvement within the overall class activities. The y-axis represents the Ch value, indicating the ratio of action transitions to the total number of actions. Based on these two metrics, each session can be categorized into one of four pedagogical types: *Lecture* ($Rt \geq 0.7$), *Practice* ($Rt \leq 0.3$), *Mixed* ($0.3 < Rt < 0.7$, $Ch < 0.4$), and *Dialogue* ($Ch \geq 0.4$).

6.2. User study

To evaluate the system's usability, its impact on teaching practices, and the accuracy of the classification model, we conducted a comprehensive user study incorporating both qualitative and quantitative feedback from users. It is important to note that the participant numbers vary across the evaluation metrics due to the distinct nature and objectives of each evaluation. The accuracy of the classification model was assessed by five experts who were familiar with the system's technical framework, ensuring a precise evaluation of its performance. In contrast, the usability and impact on teaching practices were evaluated by a larger group of 15 teachers, representing the diverse end-users of the system. The results of this study are presented in Table 8.

Accuracy of Classification: To assess the accuracy of the classification model embedded in our system, we invited five experts who were directly involved in the development of the "IRE-9" framework. These experts evaluated the classification results from 10 class recordings that had been analyzed by our system. During the evaluation process, the experts compared system-generated classifications with their own annotations. The results indicate that the system's analysis aligns closely with human perception, demonstrating high reliability.

Table 8
User study ratings (Scale: 0–5, 5 indicates high satisfaction).

Evaluation	Mean↑	Std↑	Participant
Accuracy	4.1	0.3	5
Usability	4.4	0.2	15
Enhancement	4.6	0.2	15

Usability: The usability of the system was evaluated by a mixed group of 15 teachers, which included five experienced expert teachers and 10 novice teachers, to ensure diverse perspectives. The evaluation process involved using the system over a period of two weeks, during which teachers interacted with its various features, such as dialogue analysis visualization and real-time feedback. The usability of the system interface was assessed using a standardized usability questionnaire, yielding an average score of 4.4 out of 5.

Enhancement of Teaching Practices: To understand the system's influence on improving teaching practices, we asked the same group of 15 teachers to evaluate how the system supported their instructional methods. Teachers provided feedback on specific use cases, such as how the system's analysis helped them identify overlooked student engagement patterns or refine their questioning techniques. The evaluation yielded an average score of 4.6, reflecting the positive impact of the system on enhancing classroom practices. In particular, expert teachers appreciated the real-time analysis feature, which helped them make more informed decisions during ongoing lessons, while novice teachers found the system particularly useful for post-class reflections.

6.3. Case study

To demonstrate the practical utility of the SA-DAC analysis system, we present a case study based on the dashboard visualization in Fig. 6. This classroom session combines lecturing and interactive questioning, with a Teacher Talk Ratio (Rt) of 0.58 and a Classroom Interactivity Index (Ch) of 0.43. According to the Rt-Ch pedagogical space, the session

is categorized as *Dialogue*, indicating a moderately balanced structure with frequent turn-taking between teacher and students. The bottom-right bubble chart reveals a dense cluster of teacher-led interactions in the latter half of the session. The coarse-to-fine Dialogue Act distribution (top-center pie chart) shows that I_1 and R_1 categories dominate, suggesting a pattern of factual questioning followed by student answers. The time-distribution panel (top-right) further confirms that lecturing is concentrated early in the lesson, with relatively few group-level discussions. Based on this observation, the teacher received several actionable recommendations: (1) Introduce more I_3-type open-ended prompts to foster elaborated student reasoning. (2)Increase wait time after I_3 questions to promote deeper engagement. This case illustrates how the SA-DAC system can support pedagogically meaningful reflection. Teachers can intuitively assess the quality of classroom interaction and receive targeted feedback to improve future instructional design.

7. Discussion

Although effective overall, this study encountered several limitations in model architecture, data annotation quality, and dataset scope. While the hierarchical classification approach proved generally effective, it struggled with open-ended questions, where nuanced semantics often led to ambiguity in feature extraction. This limitation suggests that current feature extraction methods may not fully capture the complexity of nuanced dialogue semantics, highlighting a need for more sophisticated semantic representations. Additionally, the quality of data annotations was affected by subjective biases among annotators, particularly in subjective categories like I_3 and R_3. Differences in annotator interpretations led to inconsistencies that impacted model performance, underscoring the importance of a standardized and objective annotation process to ensure high-quality, reliable training data.

The IRE-9 dataset, while suitable for analyzing classroom-specific interactions, has limited generalizability due to its specialized design. Furthermore, most educational datasets remain private, and research frameworks lack universal applicability (Song et al., 2022; Suresh et al., 2019), creating difficulties for cross-validation with publicly available datasets. These limitations reveal areas for improvement, guiding several directions for future work.

In future work, we plan to enhance the model's architecture by improving feature extraction methods and incorporating advanced semantic representations to better handle fine-grained categories and open-ended questions. This could help the model capture more complex dialogue patterns. To address the issue of data annotation, we will keep modifying the annotation process by improving inter-rater reliability, ensuring more consistent and reliable training data. Lastly, we aim to expand the dataset scope by including data from a wider range of grades, subjects, and classroom environments. We also plan to integrate multimodal data, such as classroom video features, to offer a more comprehensive understanding of classroom dynamics and increase the model's generalizability across different educational contexts.

8. Conclusion

Classroom dialogue analysis plays a crucial role in enhancing teaching practices and improving student learning, yet traditional manual methods are often inefficient and subjective. We propose a novel automated classroom analysis system featuring an enhanced speaker-turn aware hierarchical model-**SA-DAC**. Built on the **IRE-9** framework, our system incorporates speaker information to achieve significant improvements in dialogue classification performance. Additionally, we developed a visualization report to deepen understanding of classroom dynamics. The system's effectiveness was validated through experimental results and real-world user studies, confirming its strong potential to support and enhance educational practices.

CRediT authorship contribution statement

Linzhao Jia: Conceptualization, Methodology, Writing – original draft; **Han Sun:** Methodology, Writing – review & editing, Software; **Jialong Jiang:** Resources, Data curation, Software; **Xiaozhe Yang:** Conceptualization, Supervision, Software.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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